Tracing Functional Correctness in DLMs with Sparse Autoencoders

Anonymous ACL submission

Abstract

Prior probing of diffusion language models (DLMs) reads a hidden-state correctness signal that grows monotonically through denoising. Using DLM-Scope sparse autoencoders to trace this signal feature by feature across two DLMs and four tasks, we reach a different reading: what these probes detect largely encodes *problem difficulty*, a property of the prompt, rather than *generation-specific correctness*, a property of the output. A direct cross-generator test isolates the distinction. A probe trained on Dream-7B's L26 representation predicts Qwen-2.5-7B-Instruct's pass/fail at AUC 0.68, essentially matching its 0.72 on Dream's own labels (n=257), even though Qwen-Instruct never produced Dream's solutions. The representation therefore tracks *which prompts are hard*, not *which generations are wrong*, and the transport replicates on JSON-schema and GSM8K. A three-way per-layer scan (Qwen-Base, Qwen-Instruct, Dream-Base) shows all three reach the same peak 5-fold CV-AUC (0.71–0.73) at different layers (L19, L20, L26): diffusion training shifts the *locus* of this signal without expanding its *capacity* over instruction tuning. Feature-level tracing then exposes the temporal structure the scalar curve hides: sparse fail-vs-pass concentration is not monotonic but forms task-dependent phases. Code and structural tasks show a broad mid-denoising plateau (LLaDA-MBPP, Fisher $p < 10^{-7}$) with a late within-plateau hand-off between two persistent features, whereas reasoning tasks emerge only at the final state; the phase agrees across models on $3/4$ tasks. The difficulty reading makes a falsifiable prediction: amplifying or suppressing these features should not change outcomes, which single- and multi-feature steering confirms with $0/202$ label flips within the plateau (step 64) and only $3/202$ before it forms (step 16). Under the linear interventions we test, the sparse features behave as diagnostic correlates of prompt difficulty rather than controllable causes of correctness.

1 Introduction

Diffusion language models (DLMs) generate text by iteratively denoising a fully masked sequence over $\sim$100 steps (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Chiu and Liu (2026) trains scalar probes on hidden states and reports that aggregate correctness AUC accumulates *monotonically* through denoising, with a 0.08–0.11 gain from step 0 to the final state. That scalar view leaves three questions open: (i) *which* sparse representational components carry the failure signal, (ii) *when* during denoising they form, and (iii) whether the discovered components are merely diagnostic or also causally control correctness. Decomposing the same residual stream into named SAE features rather than a single scalar score lets us answer all three, and uncovers temporal structure that the monotonic AUC curve hides. Our central finding sharpens (iii) into a reinterpretation of what the scalar probe was reading: across architecturally matched models the signal tracks *how hard the prompt is* rather than *whether a given generation is correct*, and the steering null then follows as a prediction of that reading rather than as an isolated negative result.

Contributions.

- **Correctness probes track difficulty, not correctness.** A three-way per-layer comparison across Qwen-2.5-7B-Base, Qwen-2.5-7B-Instruct, and Dream-7B-Base shows the same peak CV-AUC ($\sim$0.72) reached at different layers (L19, L20, L26 respectively); a cross-generator test (the Dream L26 probe predicts Qwen-Instruct correctness at AUC 0.68, matching its 0.72 on Dream's own labels, though Qwen-Instruct never generated the solutions) shows the representation encodes *problem difficulty* rather than *generation-specific correctness*, and that diffusion training shifts the locus of this signal without expanding its capacity. The transport replicates across code, structural, and reasoning tasks (MBPP, JSON, GSM8K).

- **Temporal anatomy of the signal.** Feature-level tracing exposes structure the monotonic scalar curve hides: on LLaDA-8B / MBPP a 3-seed dense sweep reveals a mid-denoising plateau over steps 48–116 with a within-plateau hand-off from `f15601` to `f3892` (Fisher $p < 10^{-7}$); the peak phase is task-dependent and agrees across LLaDA-8B and Dream-7B on $3/4$ tasks.
- **Steering null as a predicted consequence.** If the features encode difficulty rather than a controllable correctness factor, amplifying or suppressing them should not change outcomes. Across 202 LLaDA-MBPP samples and 23 linear-intervention protocols (n=345 in total), the three within-plateau conditions are null at $0/202$ each; only the pre-plateau ($s16$) intervention produces a marginal $3/202 \approx 1.5\%$ effect, as predicted.

Code and figures will be released on acceptance.

2 Related Work

Discrete diffusion and masked DLMs generate text by iteratively denoising masked tokens rather than continuing left to right (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Prior probing of the same models and tasks studies aggregate hidden-state predictability and reports monotonic gains in correctness AUC through denoising. SAEs decompose residual streams into sparse named features and have been used primarily to interpret autoregressive LMs (Bricken et al., 2023; Cunningham et al., 2024; Gao et al., 2024). DLM-Scope extends this toolkit to DLMs with Top-K SAEs (Makhzani and Frey, 2014; Gao et al., 2024) trained on masked-position residuals across mask rates $t \sim U(0, 1)$ (*Mask-SAE*), keeping the full denoising trajectory in distribution (Wang et al., 2026). Recent AR SAE studies report both successful steering for refusal and instruction following (O'Brien et al., 2024; He et al., 2025) and several cautions: contrastive features may track lexical correlates, steering can degrade unrelated capabilities (Galichin et al., 2025; Korznikov et al., 2025), and SAE features need not beat random-dictionary baselines (Korznikov et al., 2026). Our held-out selection and re-selecting permutation null guard against these confounds. Tahimic and Cheng (2025) target code-correctness signals with SAEs on AR LMs, closest to our MBPP setting but without temporal trajectories. Goel et al. (2026) contrast representation evolution between diffusion and AR LMs, motivating our cross-DLM step-axis comparison. Closest to our intervention setting, Shnaidman et al. (2025) apply activation steering directly in discrete DLMs; our negative result locates one signal, the SAE-plateau difficulty feature, that resists such linear control, sharpening where steering can and cannot act. That probes may track a confound rather than the target property is a long-standing caution for probing classifiers (Belinkov, 2022); our cross-generator transport makes the specific confound, problem difficulty, measurable. We reuse the DLM-Scope SAEs to ask whether the scalar correctness trend on DLMs decomposes into temporally localised sparse features, treating the steering question as open rather than presumed causal.

3 Method

Models and SAEs. LLaDA-8B (Nie et al., 2025) (block-wise unmasking, block length 32) and Dream-7B (Ye et al., 2025) (global linear schedule) both denoise a $\leq$256-token generation region over T=128 steps conditioned on a fully visible prompt. We use the DLM-Scope L26 LLaDA-mask SAE and L23 Dream-mask SAE (trainer index 2, d_{sae}=16,384, K=160). The AR comparison (DLM-Scope L23 SAE on Qwen-2.5-7B (Team, 2024)) is a single-token reference point, not a trajectory: we encode the last-prompt-token residual once per sample as a single-step analog of the DLM's pre-generation state.

Three-way per-layer AR-DLM scan. Dream-7B is architecturally identical to Qwen-2.5-7B (28 layers, d_{model}=3584, identical attention head counts and vocab), enabling layer-aligned comparison. Throughout the paper, "Dream-Instruct" denotes the checkpoint used for all DLM generation in the cross-model cells; "Dream-Base" denotes the sibling diffusion checkpoint without instruction tuning, used only as the hidden-state source for the AR-DLM scan. For all 257 MBPP-sanitized prompts we collect the last-prompt-token hidden state at every transformer block ($L \in \{0, \ldots, 27\}$) for three models under the same chat-template prompt: Qwen-2.5-7B-Base (AR pretraining), Qwen-2.5-7B-Instruct (AR pretraining + instruction tuning), and Dream-7B-Base (AR pretraining + masked diffusion fine-tuning from the same Qwen base). We then fit per-layer logistic probes (5-fold CV, $C{=}0.01$ fixed throughout, not tuned per cell) on the Dream-Instruct fail/pass labels (so all three models predict the same generation outcomes) and report CV-AUC. This isolates two effects against the shared AR base: instruction tuning (Qwen-Base $\rightarrow$ Qwen-Instruct) and diffusion fine-tuning (Qwen-Base $\rightarrow$ Dream-Base).

Data. We reuse cached hidden states and functional-correctness labels from Chiu and Liu (2026): MBPP-sanitized (257) (Austin et al., 2021b), JSON schema (272), GSM8K (1,319) (Cobbe et al., 2021), and ARC-Challenge (1,172) (Clark et al., 2018), generated with seed 0 and $T{=}128$ steps. Hidden states are mean-pooled over four equal-length generation regions at seven checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$.

Per-step SAE encoding. At each (model, step, layer) cell we take the region-mean hidden state $h \in \mathbb{R}^d$ per sample and encode it: $z = \mathrm{TopK}\big(\mathrm{ReLU}(W_{\mathrm{enc}}(h - b_{\mathrm{dec}}) + b_{\mathrm{enc}})\big) \in \mathbb{R}^{d_{\mathrm{sae}}}$, with $W_{\mathrm{enc}} \in \mathbb{R}^{d_{\mathrm{sae}} \times d}$, $W_{\mathrm{dec}} \in \mathbb{R}^{d \times d_{\mathrm{sae}}}$, $b_{\mathrm{enc}} \in \mathbb{R}^{d_{\mathrm{sae}}}$, $b_{\mathrm{dec}} \in \mathbb{R}^d$, and TopK keeps the $K{=}160$ largest entries. Feature i is *active* on a sample iff $z_i > 0$.

Fail-vs-pass enrichment. Let $\mathcal{F}, \mathcal{P}$ be the fail and pass sets at a cell. Feature i's enrichment is the difference of active rates, $\mathrm{enr}(i) = \frac{1}{|\mathcal{F}|}\sum_{s\in\mathcal{F}} \mathbb{1}[z_i^{(s)} > 0] - \frac{1}{|\mathcal{P}|}\sum_{s\in\mathcal{P}} \mathbb{1}[z_i^{(s)} > 0]$. We use the top-20 most fail-enriched features as the analysis subset per cell.

Cluster discriminability with permutation null. We run KMeans on the fail samples for $K \in \{2, 3, 4, 5\}$ using the top-20 features (continuous activations) and report the silhouette of the best K. The permutation null shuffles labels $1000\times$, *re-selects* the top-20 features on each shuffled label set, then recomputes silhouette, controlling for the post-hoc feature selection. We report $p = \Pr[\mathrm{silhouette}_{\mathrm{perm}} \geq \mathrm{silhouette}_{\mathrm{obs}}]$. The silhouette is computed on fails only and measures sub-structure within failures, not fail-vs-pass separability directly. We therefore interpret it as *concentration of failure modes* in a selected sparse subspace, and cross-check fail/pass separability with supervised AUC in the same subspace under per-fold selection (Appendix E).

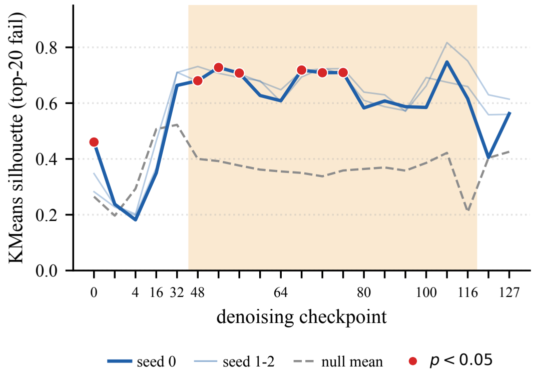

Figure 1: LLaDA-8B / MBPP: KMeans silhouette on the top-20 fail-enriched features vs. permutation null mean across denoising steps. All 21 checkpoints (anchors $\{0, 1, 4, 16, 32\}$, early plateau $\{48–80\}$, late plateau $\{84–124\}$, final 127) are from a single new-prompt generation run; faint lines overlay two additional generation seeds (Appendix I). Shaded region marks the mid-denoising plateau $[48, 116]$ where most steps reach $p < 0.05$; step 64 sits inside the plateau as a local dip, not a sharp peak.

Steering. We write *fail→pass* for a sample whose label flips from fail to pass after intervention, and *pass→fail* for the reverse pass-side regression. Given a target feature $i^\star$ and strength α, a forward hook on the SAE layer's transformer block re-encodes the residual at every step $t \geq t_0$, reads $z_{i^\star}$, and in-place subtracts $\alpha \cdot z_{i^\star} \cdot W_{\mathrm{dec}}[:, i^\star]$. $\alpha < 0$ amplifies the feature (reverse intervention); for multi-feature steering we sum contributions over a set of indices.

4 Results

Mid-denoising plateau. Figure 1 reports a 21-step trajectory for LLaDA-MBPP across three seeds: anchors $\{0, 1, 4, 16, 32\}$, early plateau $\{48, \ldots, 80\}$, late plateau $\{84, \ldots, 124\}$, and final state 127. Rather than a sharp peak at step 64, the trajectory shows a broad plateau over steps 48–116: the signal-to-null gap stays in $[+0.19, +0.41]$ and only collapses at step 124. `f15601` is the top-1 fail-enriched feature at $7/9$ early-plateau dense steps and remains dominant through step 100; `f3892` replaces it at steps 108–116, indicating a within-plateau hand-off rather than feature swapping outside the plateau. At steps 0–16 the top fail feature instead changes every step (Appendix A), consistent with the residual stream not yet encoding committed-token decisions when most positions are still masked. Figure 3 contrasts the per-sub-step gap on LLaDA-MBPP and Dream-MBPP: LLaDA-MBPP has 6–9 of 14 dense sub-steps at $p < 0.05$

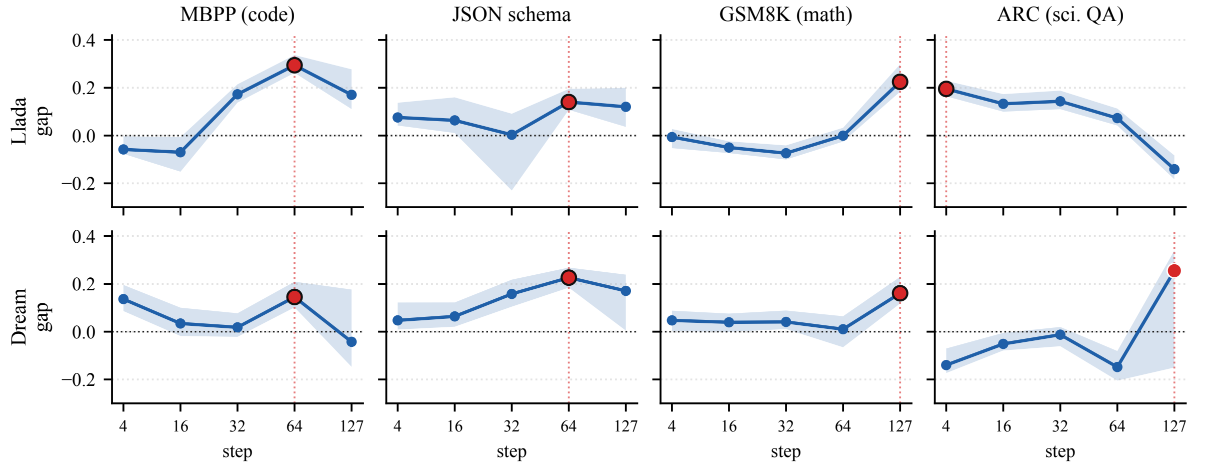

Figure 2: Cross-model × cross-task trajectory grid. Each panel plots signal-to-null gap (observed KMeans silhouette minus permutation null mean) across denoising steps for one (model, task) cell; shaded band = 95% bootstrap CI over fail samples; red marker = peak gap step, filled when its CI excludes zero. Code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64) on both DLMs. Math (GSM8K), and on Dream science QA (ARC), peak at late denoising (step 127). The peak step shows task-level agreement across LLaDA-8B and Dream-7B on three of four tasks; only ARC disagrees (LLaDA peak step 4, Dream step 127). We read this as a phase *tendency*: individual steps are mostly not significant after Holm-Bonferroni (§4), and only the aggregated trajectory tests reach significance.

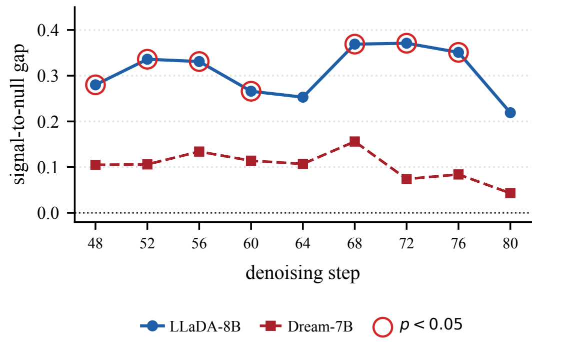

Figure 3: Dense step sweep (48–80 every 4 steps) on LLaDA-MBPP and Dream-MBPP, seed 0. Hollow red rings mark LLaDA sub-steps with permutation $p < 0.05$ (7 of 9); no Dream sub-step reaches the same threshold under the matched null. The mid-denoising plateau on LLaDA-MBPP aggregates to Fisher $p < 10^{-7}$ on every seed.

per seed (Fisher combined $p = 5.99 \times 10^{-10}$ for seed 0), while Dream-MBPP shows the same plateau shape with a much lower concentration (gap never exceeds $+0.16$, no sub-step reaches $p < 0.05$). The LLaDA × MBPP combination is therefore a model-specific point where the residual stream factors fail-vs-pass into a small set of sparse features rather than a generic property of code generation. The raw-probe AUC in Chiu and Liu (2026) *monotonically* increases over the same steps (LLaDA-MBPP, $\sim$0.75 at $s0$ to $\sim$0.84 at $s64$), framing denoising as gradual scalar sharpening, whereas the feature-level plateau and the late hand-off are invisible to

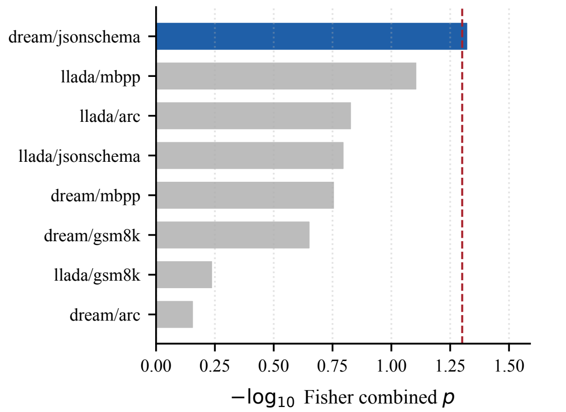

Figure 4: Fisher's combined p across the sparse 5-step grid per (DLM, task) cell; red dashed line at $p = 0.05$. LLaDA-MBPP and Dream-JSON reach aggregated significance; reasoning tasks do not, and LLaDA-ARC remains an outlier driven by prompt-boundary failure modes.

scalar probing.

Cross-model × cross-task phase agreement. Figure 2 reports the full 2×4 trajectory grid; Appendix C lists step-64 numbers. Code and structural cells (MBPP, JSON on both DLMs) peak mid-denoising at step 64, reasoning cells (GSM8K on both, Dream-ARC) peak at step 127, and LLaDA-ARC is an outlier peaking at step 4. Three of four tasks therefore share a peak step across LLaDA and Dream, mirroring the structural-vs-reasoning emergence split of Chiu and Liu (2026) as a peak position rather than a continuous AUC curve. No cell survives Holm-Bonferroni across 40 sparse-grid tests (8 cells $\times$ steps $\{4, 16, 32, 64, 127\}$, dropping the two fully-masked steps $\{0, 1\}$ where the null is dominated by mask geometry), but two cells reach aggregated significance under Fisher's combined probability test on each cell's sparse 5-step trajectory (Figure 4): the LLaDA-MBPP dense plateau (Fisher $p < 10^{-7}$ per seed) and Dream-JSON ($p = 0.047$); the LLaDA-MBPP plateau is also L26-specific under a cross-layer scan (Appendices F, G, H). The ARC anomaly fits this picture: LLaDA-ARC fails near the prompt boundary so its early-step cluster captures prompt-attention failure modes rather than reasoning-stage failure, accounting for the step-4 peak versus Dream-ARC's step-127 peak.

Top-feature drift and persistence. On LLaDA-8B / MBPP, pairwise Jaccard of the top-20 sets is high among steps 0–4 (≥ 0.48), moderate between 32 and 64 (0.29, the plateau regime), and falls to 0.14 between 64 and 127. The dense sweep (Appendix F) ranks `f15601` top-1 at 7 of 9 dense steps in $[48, 80]$, supporting a persistent statistical marker inside the plateau and a concentration effect in sparse feature space rather than a monotonic increase in correctness information. Selecting the top-20 on held-out folds rather than in-sample leaves all nine dense sub-steps significant (Appendix K), so the plateau does not depend on in-sample feature selection. A supervised cross-check confirms this picture: top-20 SAE features recover 85–95% of the raw-probe AUC at step 64, with raw probing winning in 10 of 12 cells (the two SAE wins, Qwen-MBPP and Dream-ARC, are cells where the raw probe is itself near chance), so the sparse subspace captures most yet not all of the dense-residual signal (Appendix E, Figure 9). This 5–15% residual motivates testing raw-residual directions that occupy the full hidden state (§4); their null result rules out the sparse-subspace restriction as the explanation, leaving the difficulty reading as the parsimonious account of why neither sparse nor dense linear directions move the outcome.

Steering result. We test whether the cluster-level signature can be directly controlled on LLaDA-MBPP, the only cell with cluster significance. The steering generation is a separate seed-0 decoding run yielding 105 pass and 152 fail (versus the dense sweep's $115/142$, Appendix I); we report all 152 fails and 50 pass-side regression controls across four interventions (Table 1). Suppressing `f15601`

condition	from	fail→pass	pass→fail
suppress `f15601`	s64	0/152	0/50
suppress `f15601`	s16	**2/152**	**1/50**
suppress top-5	s64	0/152	0/50
reverse `f15601`	s64	0/152	0/50

Table 1: Full steering intervention summary on LLaDA-8B / MBPP; entries show flips / evaluated cases over all 152 fail samples (126 cluster-1 + 26 cluster-0) and 50 pass-side regression controls. Suppress `f15601` or the top-5 features at step 64 (within the plateau) is fully null at $0/202$; suppress `f15601` from step 16 (before plateau formation) produces a small $3/202$ effect (1.5%, both fail→pass conversions occur on cluster-1 cases). Reverse intervention ($\alpha = -5$) is null. The $0/202$ plateau null has a 95% Clopper-Pearson upper bound of 1.8% (1.1% for the full $0/345$ sweep), so the protocol is powered to exclude within-plateau flip rates above $\sim$2%.

or the top-5 fail features within the plateau (step 64), and the reverse intervention ($\alpha = -5$), each yield $0/202$ label flips, as does a counterfactual replacement that subtracts the top-5 fail-enriched and adds the top-5 pass-enriched directions. A broader sweep of 23 linear protocols on small n (Appendix J) covering both sparse SAE directions (17 configs, magnitudes up to $\alpha = 50$) and raw residual directions (mean-difference, PCA, Fisher LDA; 6 configs) yields $0/345$ fail→pass total. Only step-16 intervention, before plateau formation, produces a marginal effect ($3/202$). The null is not an artifact of a silent perturbation: at step 64 the SAE-layer residual edit reaches $\|\Delta h\|/\|h\| = 4.4\%$ for single-feature suppression and 14.8% for top-5 suppression, and the largest raw mean-difference protocol ($\alpha = 50$) drives $\|\Delta h\|$ to $\sim$17$\times$ the residual norm, well beyond the magnitudes used in successful AR steering studies, yet still flips $0/15$ cluster-1 fails. Nor is it an artifact of intervening at a single layer: jointly suppressing each layer's top-1 fail feature at L11, L16, L26, and L30 *simultaneously* (hooks registered together, $\alpha = 5$, $t \geq 64$) also flips $0/40$ fails and regresses $0/20$ pass controls, so the multi-layer write that distinguishes the AR-DLM representations (§4) is not recovered by a joint linear intervention either. The target feature is reliably active during every intervention window, so a $0/202$ result at the plateau is consistent with functional correctness being distributively encoded rather than carried by the single direction the SAE plateau makes salient, which is the difficulty signal the cross-generator test isolates below (§5). A zero-out smoke test ($h \leftarrow 0$) confirms the hook propagates: it garbles trailing tokens but leaves the early-committed function body intact, consistent

feature	dominant context (F/P)	top quotation
`f15601`	complex-math doc-string (25/5)	*"nth octagonal number..."*
`f3892`	`:param`/`:return` (15/15)	"`:param tuple_list:` *list*"
`f11265`	broken-syntax / incomplete (17/13)	*"dp[n == 0] = [1 \n"*

Table 2: Activation-maximising contexts (top-30 *(sample, position)* entries per feature) on LLaDA-8B / MBPP, step 64. `f15601` concentrates on complex-math docstrings, fail/pass split 25/5. `f11265` fires on already broken syntax. Full table including `f5561`, `f2144` in Appendix L.

with LLaDA committing blocks 0–3 by step $\sim$48.

What `f15601` actually encodes. To turn the statistical marker into an interpretable variable we re-run LLaDA at step 64 on all 257 MBPP samples, encode every generation position through the L26 Mask-SAE, and record the top-30 *(sample, position)* entries per target feature. Table 2 shows the result for three top features (full table in Appendix L). The mid-plateau peak feature `f15601` concentrates on docstring positions containing dense mathematical specifications (Eulerian / octagonal / hexagonal / Babylonian formulae): in its top-30 activations the fail/pass split is 25/5, but the context is the *problem statement* rather than any concrete failure mode in the model's output. This makes `f15601` a *task-difficulty correlate* that fires upstream of any specific failure: ablating it removes the model's encoding of "this problem is hard," which is necessary for predicting failure but not sufficient to change the generation trajectory. The late-plateau replacement `f11265` fires on contexts that already contain visible syntactic defects in the partially-generated code (top quotation "`dp[n == 0] = [1 \n dp = [0]`"), suggesting the within-plateau hand-off from `f15601` to `f3892` (and the late drift toward `f11265`) reflects a transition from *difficulty signalling* to *outcome signalling* as denoising completes.

L26 is the unique cluster-significant layer. To check that L26 is not an arbitrary choice, we re-run the silhouette-with-permutation-null pipeline at the four available DLM-Scope LLaDA SAE layers $\{11, 16, 26, 30\}$ at plateau steps 64 and 68 (Figure 5). L26 is the unique peak: step 68 at L26 gives gap $+0.357$ with permutation $p = 0.006$, while L11, L16, and L30 give gaps below $+0.21$ and none reach $p < 0.05$. Moreover `f15601` appears as top-1 fail-enriched feature only at L26; different layers select different atoms (Appendix G). The persistent-feature claim is therefore L26-specific

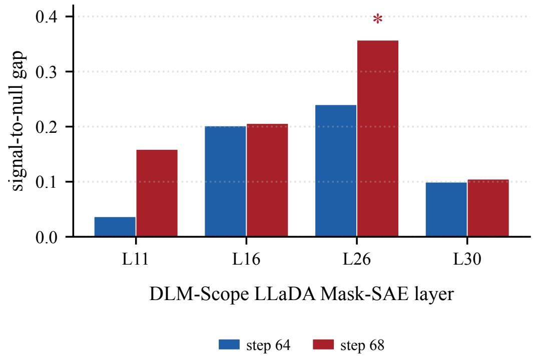

Figure 5: LLaDA-MBPP signal-to-null gap across SAE layers at plateau steps 64 and 68. L26 is the unique layer where the gap reaches permutation $p < 0.05$ (step 68, gap $+0.357$). Top-1 fail-enriched feature differs by layer; `f15601` appears only at L26.

model	peak L	peak AUC	L27 AUC	$\cos_{27}$
Qwen-Base (AR)	19	0.712±.06	0.631	1.00
Qwen-Instruct	20	0.730±.07	0.708	0.16
Dream-Base (DLM)	26	0.715±.04	0.696	0.12

Table 3: Per-layer CV-AUC peaks on MBPP-Dream labels (n=257, $\pm$ per-fold std). The three peaks lie within one standard deviation of one another, so peak capacity is statistically indistinguishable while the locus shifts from L19/20 to L26. At L27 both post-trained models diverge from Qwen-Base ($\cos < 0.17$); the rewrite is not diffusion-specific.

rather than a generic mid-layer property, supporting our choice of L26 as the analysis layer. Repeating the suppression protocol of §4 at the three non-L26 layers using each layer's top-1 fail feature ($\alpha = 5$, $t \geq 64$, $n = 60$ per layer) yields 0/60 flips at each layer: the within-plateau steering null generalises across the layer span tested, so the difficulty reading is not L26-specific (Appendix G).

Diffusion shifts the locus of correctness encoding without expanding capacity. Figure 6 shows the per-layer CV-AUC for the three architecturally matched 7B models on the same prompts and Dream-Instruct labels (n=257); Table 3 summarises the three peaks. Three observations: (i) the three models reach equivalent peak AUC (0.712, 0.730, 0.715, all within one per-fold std of each other; Table 3) but at different layers (L_{19}, L_{20}, L_{26}); (ii) diffusion training does not exceed instruction tuning at peak; (iii) post-training rewrites the final layer regardless of objective: at L27 both Qwen-Instruct and Dream-Base diverge from Qwen-Base (cos 0.16 and 0.12 respectively) and from each other (cos 0.13). The L27 collapse of Qwen-Base AUC (from 0.71 at L19 to 0.63) is recovered by both post-training procedures (0.71

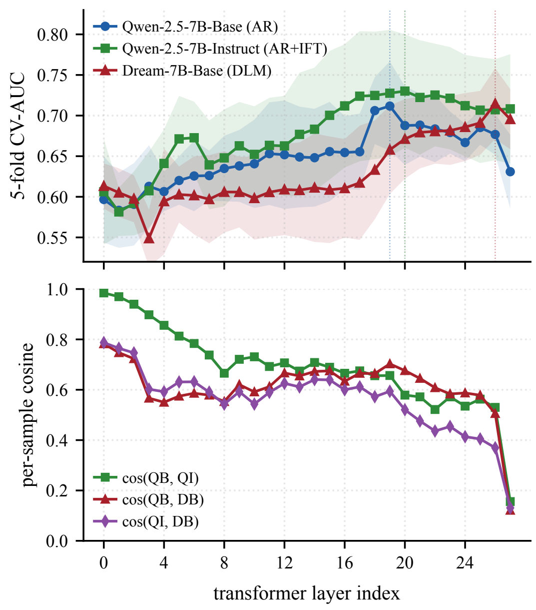

Figure 6: Per-layer 5-fold CV-AUC for predicting Dream-Instruct MBPP correctness from the last-prompt-token residual at each transformer block (top), and per-sample cosine similarity between the three models (bottom). All three models are architecturally identical (28 layers, d=3584, same vocab). Shaded bands are $\pm$ per-fold std. Qwen-Base peaks at L19 (0.71), Qwen-Instruct at L20 (0.73), Dream-Base at L26 (0.72); the peaks lie within one std (overlapping bands), so capacity is shared while the locus shifts. At L27, both post-trained models diverge sharply from Qwen-Base (cos$<$0.16) and from each other (cos 0.13), indicating post-training rewrites the final layer regardless of objective.

for Qwen-Instruct, 0.70 for Dream-Base), so the late-layer divergence is not diffusion-specific. Critically, Qwen-Instruct does not generate the Dream solutions, yet matches Dream's L26 peak with an L20 peak of comparable AUC: this implies the per-layer AUC reflects *problem-difficulty encoding*, which is shared across architectures and post-training procedures, rather than DLM-specific generation-correctness encoding.

5 Discussion

Why linear steering fails. The three-way scan supplies a candidate mechanism: Dream's L26 representation encodes *which prompts are hard*, not *which generations are wrong*, and the same difficulty reading is the most parsimonious account of LLaDA's L26 SAE plateau, which is what steering targets. A linear suppression at L26 therefore subtracts a difficulty estimate from the residual; the underlying generation trajectory, decided by token-level decoding decisions throughout denoising, is unaffected.

Cross-generator test. We test the difficulty reading directly by re-running MBPP-sanitized with Qwen-2.5-7B-Instruct (pass rate $190/257 = 74\%$, per-sample pass/fail agreement with Dream-Instruct 0.642) and asking how well *Dream*'s L26 hidden states predict *Qwen-Instruct*'s correctness on the same problems. Under 5-fold logistic CV with $C = 0.01$, the Dream L26 probe predicts Qwen-Instruct labels at AUC 0.681, essentially matching its 0.715 on Dream's own labels (diff -0.033). The L26 probe therefore transports across generators with little AUC loss, suggesting much of the predictive signal is shared problem difficulty rather than Dream-specific self-correctness. This replicates on two further tasks (Appendix N): on JSON-schema a probe on Dream's representation predicts Qwen-Instruct's correctness at least as well as Dream's own at every upper layer (best 0.771 on Qwen vs 0.744 on Dream; agreement 0.783), and on GSM8K it predicts Qwen-Instruct *better* than Dream itself (0.719 on Qwen vs 0.651 on Dream), so the transport holds across code, structural, and reasoning tasks rather than being specific to MBPP. This reading aligns with the three steering signatures: the full $0/345$ flip rate across sparse and raw-residual directions, the marginal $3/202$ effect only at step 16 before any token commits, and Qwen-Instruct's higher L20 AUC despite not being the generating model. A capacity-limited DLM control protocol therefore needs features that encode commitments rather than difficulty, plausibly at earlier layers or denoising steps, or non-linear interventions that re-route the trajectory rather than scaling a single direction.

Difficulty reading and the task-dependent phase. The difficulty reading also organises the cross-task peak pattern of Figure 2. A difficulty signal is most predictive once the prompt is encoded but before the answer commits, exactly the mid-denoising window where code and structural tasks plateau: the hardness of a spec is fixed by the docstring and readable throughout the commit phase, matching `f15601`'s concentration on dense-math docstrings (Table 2). Reasoning tasks differ because their difficulty is inseparable from the answer the model is still constructing, so the fail-vs-pass signal cannot concentrate until the final state, where GSM8K and Dream-ARC peak. On this reading the structural-vs-reasoning split that Chiu and Liu (2026) report as a continuous AUC gap is the temporal shadow of a single mechanism, namely *when* prompt difficulty becomes estimable from the residual, rather than two separate correctness signals.

Relation to AR SAE steering successes. Recent AR SAE work reports successful steering for refusal (O'Brien et al., 2024) and instruction following (He et al., 2025); Tahimic and Cheng (2025) steer code-correctness signals at the SAE layer on AR LMs. The difficulty-encoding mechanism we identify is consistent with that body of work rather than in tension with it: refusal and instruction following are stylistic axes not predictable from problem difficulty, so an AR SAE feature can causally affect them without overlapping a difficulty signal, whereas code correctness is the harder target where Tahimic and Cheng (2025) achieves a smaller effect. We therefore read the DLM steering null as a sharpened instance of a general SAE-correctness limitation rather than a DLM-specific failure mode; the per-layer scan further suggests that for DLMs the steering layer should track the model's self-correctness peak rather than the SAE plateau.

A methodological caution for correctness probing. The cross-generator transport has a consequence beyond DLMs: a probe that predicts correctness with above-chance AUC does not establish that the model represents *its own* correctness, because problem difficulty is confounded with outcome whenever harder prompts fail more often. On MBPP this confound is large enough to carry most of the predictive signal, since the Dream L26 probe loses only 0.033 AUC when transferred to a model that did not generate the labelled outputs. A correctness probe is thus best read as an upper bound that mixes a prompt-difficulty term, transportable across generators, with a generation-specific term, which is the part a control protocol can actually move. Two checks separate the terms without new training: transport to an independent generator (AUC retained is the difficulty floor, AUC lost the generation-specific component) and re-fitting within difficulty-matched strata (Appendices N, P); both indicate difficulty dominates the signal across our tasks. Practically, steering should target commitment-level features at the self-correctness peak (L19–L20 for the AR siblings) rather than the difficulty-dominated SAE plateau, and reported correctness-probe AUCs should be discounted by an estimated difficulty floor before being read as evidence of self-monitoring.

Future directions. Three concrete avenues follow from the present results. *Commitment-level features.* The actmax contexts show `f11265` as the late-plateau feature that fires on *already broken* partial code (Table 2); a Mask-SAE trained on Unmask positions (already revealed tokens) and selected on the late hand-off step, or a gradient-aware dictionary that prioritises downstream influence (Olmo et al., 2025), might surface a similar outcome-side feature whose ablation is closer to causal. *Multi-layer joint steering.* The cross-layer specificity test (Figure 5) and the AR-DLM per-layer scan (Figure 6) both indicate that the correctness-related representation is layer-distributed; a single-layer linear intervention cannot recover the multi-layer write that distinguishes Dream from Qwen-Instruct. *Pre-commit interventions.* The marginal $3/202$ flips occur only when the intervention starts at step 16, i.e. before any block commits; targeting earlier steps with appropriately matched directions is the most direct way to test the commitment-timing constraint.

6 Conclusion

Tracing functional-correctness signal feature by feature reinterprets what hidden-state probes read in DLMs: the signal largely encodes problem difficulty, a prompt property, rather than generation-specific correctness, a reading that rests on cross-generator transport and predicts the steering null. Feature-level tracing also reveals temporal structure scalar probes miss: sparse failure markers form task-dependent phases, code and structural tasks on a mid-denoising plateau and reasoning tasks peaking at the final state (Fisher $p < 10^{-7}$ on LLaDA-MBPP, agreement on $3/4$ tasks), with a late `f15601`→`f3892` hand-off. Linear interventions flip $0/202$ within the plateau (and $0/40$ under joint multi-layer suppression); since Qwen-Instruct, which never generated the Dream solutions, matches Dream's L26 peak, the representation tracks difficulty inherited from the AR base, and causal control needs commitment-level features at earlier layers, not amplification of the difficulty signal.

Limitations

SAE coverage. All findings depend on a single DLM-Scope Mask-SAE per model (L26 for LLaDA, L23 for Dream), trained with mask rate $t \sim U(0, 1)$ on masked-position residuals. The full denoising trajectory is therefore in distribution, but features extracted at other layers, the Unmask-SAE trained on already-revealed positions, or higher-capacity dictionaries ($K\!>\!160$) may surface different failure correlates; re-encoding through $k \in \{50, 160, 520\}$ dictionaries (Appendix Q) shows the plateau significance is dictionary-robust while the exact top atom is sparsity-dependent. Early-step instability (steps 0–16) reflects the residual stream not yet encoding committed-token decisions rather than SAE distribution shift.

Feature semantics. We identify persistent dictionary atoms such as `f15601` and `f3892` by statistical enrichment and temporal persistence, not by human-validated semantic labels. The present experiments therefore support a claim about when sparse failure markers concentrate; Appendix L characterises four of them via activation-maximising contexts, but full mechanistic naming would require systematic intervention beyond this protocol.

Permutation test interpretation. Our permutation null re-selects the top-20 fail features on each shuffled label set before recomputing silhouette. This is conservative (any post-hoc selection procedure that could find clusters on random labels is included in the null), and it inflates the null mean noticeably (e.g. 0.36 for LLaDA-8B / MBPP at step 64). A consequence is that only two of twelve cells reach $p < 0.05$ at an individual step on the sparse 7-grid even though the DLM grid shows systematic plateau structure. Aggregated tests partially mitigate this: the 14 dense LLaDA-MBPP plateau sub-steps yield Fisher combined $p\!<\!10^{-7}$ on every seed and held-out feature selection (Appendix K) reaches $p\!<\!0.05$ on all 9 dense sub-steps; we present cross-model claims as plateau-shape claims rather than per-cell significance.

Steering coverage. Our intervention sweep tests four denoising windows, two feature-set sizes (single and top-5), and one magnitude pair ($\alpha \in \{+5, -5\}$) at a single SAE layer per DLM (L26 for LLaDA, L23 for Dream). We add a joint multi-layer intervention (suppressing each of L11/L16/L26/L30's top-1 fail feature at once), which is also null ($0/40$), but we do not test non-linear projections or feature-selection strategies that use the per-step top-1 (where the top fail feature differs from `f15601`). Effect-size diagnostics (Appendix J) confirm the intervention reaches $\|\Delta h\|/\|h\| \in [4.4\%, 14.8\%]$ at the SAE-layer residual, so the within-plateau null is not because the perturbation was silent; rather, a $0/202$ result on this protocol is narrower than a negative result on all possible linear-feature interventions.

Cluster K and choice of metric. We use silhouette on KMeans with $K \in \{2, 3, 4, 5\}$ and report the best K per cell. On LLaDA-8B / MBPP the best K is 2 at every step; on Dream / MBPP step 64 the best K is 3 with one small (size-26) sub-cluster. Appendix K reports top-N and clustering-method sensitivity: silhouette and significance are robust across $N \in \{10, 20, 30, 50\}$ at step 68 (the plateau peak) and across KMeans/Agglomerative/HDBSCAN. A labels-hidden $K\!=\!2$ test on the full fail+pass set yields ARI ≈ 0, confirming the silhouette measures within-fail sub-structure rather than fail/pass separability; supervised AUC in Appendix E reports the latter.

Cross-model comparison. LLaDA uses blockwise unmasking with block length 32, whereas Dream uses a global linear schedule. At a matched checkpoint step, the spatial distribution of unmasked tokens differs between the models, so step-axis comparisons in Figure 2 reflect approximate denoising-progress alignments rather than identical model states. The unmasked *fraction*, however, matches closely at every checkpoint ($\leq 1.6\%$ apart, both $\sim\!50\%$ at step 64; Appendix R), so the comparison is aligned in denoising progress even though the per-position commit order differs. Our AR baseline (Qwen-2.5-7B) uses last-prompt-token hidden states; this is the natural single-token analog of the DLM's pre-generation state but is not a temporal trajectory.

Task coverage. Functional correctness is verified across four tasks (MBPP, JSON schema, GSM8K, ARC), but this still covers only a small slice of DLM use cases. Generalising the task-dependent peak finding to other open-ended tasks (instruction following, dialog, multi-turn reasoning) requires task-appropriate functional metrics and additional DLMs.

Sample size and seed dependence. Cluster sample sizes are limited by the number of fail cases per (model, dataset) cell. MBPP has $\sim\!140$ fails per DLM, while GSM8K has fewer than 250 each. A three-seed sensitivity check on LLaDA-MBPP (Appendix I) shows pass/fail counts vary by at most ± 3 samples and the plateau structure is preserved across seeds. The temporal-oscillation phenomenon noted in concurrent DLM work (Wang et al., 2025) would primarily affect borderline samples, a pattern consistent with our seed-stability check.

Difficulty vs. correctness disentanglement. Our central claim, that the L26 representation tracks problem difficulty rather than generation-specific correctness, is supported by two complementary instruments: cross-generator transport (§5, Appendix N) and prompt-difficulty stratification (Appendix P). Both are informative but neither is complete. The retained AUC under transport (0.68 of 0.72 on MBPP, and a slight *gain* on JSON) bounds the difficulty floor from below only to the extent that Qwen-Instruct and Dream share a difficulty ordering over these prompts; if the two models found systematically different prompts hard, transport would understate the generation-specific component and our claim would be too strong. The residual AUC gap on MBPP (0.033) is consistent with a small but non-zero generation-specific term, so the honest statement is that difficulty *dominates* the signal, not that correctness is absent from it; the stratification instrument is consistent with this, removing only 0.026–0.036 AUC under prompt-length matching. Transport holds across four datasets (MBPP, JSON, GSM8K, and held-out HumanEval) and three generators (Dream-Instruct, Qwen-Instruct, Llama-Instruct), which triangulates the shared difficulty ordering rather than relying on one model pair (Appendix N); the residual generation-specific term is what remains for a control protocol to move. Finally, the LLaDA L26 SAE plateau is linked to the same difficulty reading by parsimony and by the shared L26 locus, not by a cross-generator test on LLaDA itself, which would require an architecturally matched AR sibling for LLaDA that we do not have.

Causal interpretation. Even with the negative steering result, our conclusion that "`f15601` is a diagnostic correlate rather than a directly controllable cause under our protocol" is interpretation-dependent. Two converging checks support it beyond plain suppression: a counterfactual replacement that swaps in paired pass-cluster directions ($0/202$, §4) and a gradient attribution in which `f15601` ranks 220th of 491 active features for influence on the output (Appendix S). A fully mechanistic account would still require tracing the multi-layer write that distinguishes the AR-DLM representations, which we leave to future work.

Broader impact. Our main finding is a caution against over-reading correctness probes, so its direct use is defensive: it discourages treating a difficulty-correlated feature as a reliable lever for behaviour change. Two risks attend the tooling. Steering hooks like ours could be repurposed to search for failure-inducing inputs rather than to improve reliability, and "difficulty" features estimated this way should not be reused as proxies for protected attributes when similar analyses are applied to user-generated content, since prompt difficulty and demographic correlates can be entangled in open-domain data.

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A Top-Feature Persistence Across Steps

Figure 7 shows the pairwise Jaccard of top-20 fail features between denoising steps and the enrichment trajectory of three illustrative features for LLaDA-8B / MBPP. `f15601` is the only feature among the top-20 at all three peak-or-later steps (32, 64, 127), and is the top-1 fail-enriched feature at both step 32 ($+0.391$ enrichment) and step 64 ($+0.385$); on the sparse 7-grid only step 64 reaches $p < 0.05$ (Figure 1 reports the dense sweep that locates the plateau).

B Qualitative Cluster Inspection

Table 4 shows illustrative LLaDA-MBPP failures from the two step-64 fail clusters. We re-generated the first ten fail samples from each cluster with the baseline LLaDA decoding setup and re-ran the MBPP tests. The examples are not intended as human-validated feature labels; they only show that both sparse clusters contain concrete functional failures, ranging from malformed code to plausible but semantically wrong implementations.

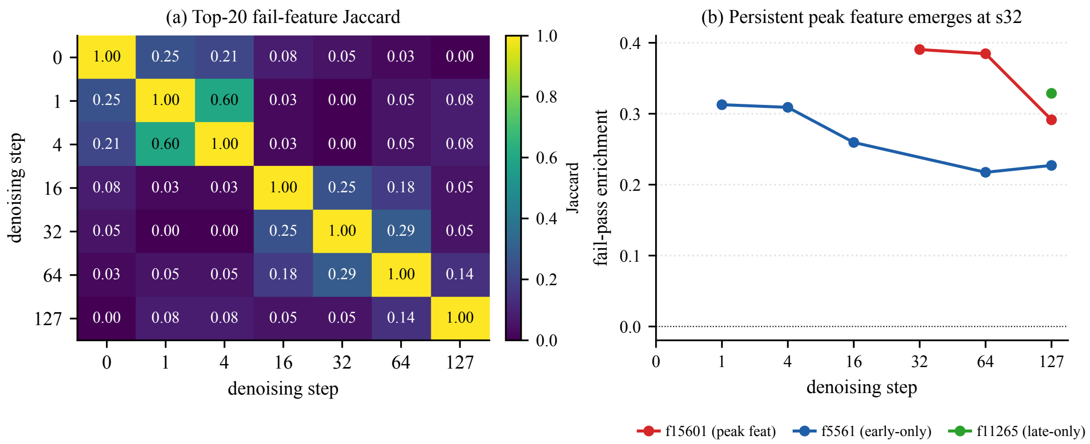

Figure 7: Left: pairwise Jaccard of top-20 fail features between denoising steps. Right: enrichment trajectory of `f15601` (persistent peak), `f5561` (early only), and `f11265` (final only).

#	characteristic features	illustrative failures
0	`f11404`, `f9798`, `f189`, `f496`, `f8825`	*closest smaller number*: returns n for small inputs instead of $n-1$, failing assertions. *unique element in sorted array*: malformed pointer-update code, syntax error.
1	`f189`, `f492`, `f3320`, `f11404`, `f15601`	*remove first/last occurrence*: deletes only boundary characters, missing interior occurrences. *lowercase underscore regex*: uses `re.compile` without importing `re`, `NameError`.

Table 4: Illustrative generated failures from the two LLaDA-MBPP step-64 fail clusters. Error-type counts over 10 regenerated failures per cluster: cluster 0 has 3 assertion / 2 syntax / 4 name / 1 value error; cluster 1 has 5 assertion / 4 name / 1 type error.

C Step-64 Cross-grid Snapshot

Table 5 summarises silhouette + permutation p across the 4 datasets $\times$ 3 models grid: DLM step-64 plateau point plus the single-step Qwen AR reference, complementing the trajectory-shape claim in Figure 2.

dataset	LLaDA	Dream	Qwen
mbpp	0.66 / **0.033**	0.50 / 0.128	0.16 / 0.106
json	0.58 / 0.195	0.77 / 0.135	0.22 / **0.007**
gsm8k	0.38 / 0.432	0.29 / 0.449	0.15 / 0.315
arc	0.43 / 0.312	0.18 / 0.882	0.13 / 0.343

Table 5: Silhouette / permutation p per condition: DLM step-64 for LLaDA / Dream, single-step AR reference for Qwen; bold $p < 0.05$. Full DLM trajectory in Figure 2.

D Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen\_length}/32)$ steps per block. Within each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i, the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with #### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 8.

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant that answers in JSON. Here's the JSON schema you must adhere to:\n <schema>\n {schema} \n </schema>\n user\n {user input}\n assistant\n ``'json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem step by step. End your answer with #### followed by the final numeric answer. im_end\n im_start user\n {question} im_end\n im_start assistant\n
```

Figure 8: Schematic prompt examples after each model's chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (≤ 256 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the residual stream at the SAE layer. The generation region is partitioned into 4 equal-length position regions and mean-pooled within each region, producing a d-dimensional vector per region (4096 for LLaDA, 3584 for Dream and Qwen). The region-mean is passed through the DLM-Scope SAE for feature analysis (§3).

Steering hook implementation. The steering hook is registered on the SAE-layer transformer block via `register_forward_hook`. The hook receives the tuple output, re-encodes `output[0]` through the SAE in-place, computes $\alpha \cdot z_{i^\star} \cdot W_{\text{dec}}[:, i^\star]$ and subtracts the contribution. A counter and per-step diagnostic prints confirm the hook fires $T{=}128$ times per generation (once per denoising step) and that the modification magnitude $\|\Delta h\|$ is non-zero.

Computational resources. All generation, encoding, and steering experiments use NVIDIA A100 (80GB) GPUs on Modal. Models are loaded in bfloat16; SAE weights cast to bfloat16 for the steering hook. Total compute: approximately 40 A100-hours across all reported experiments.

Reproducibility and determinism. Generation uses seed 0; both LLaDA and Dream commit the highest-confidence masked positions deterministically, so the fail/pass partition per cell is reproducible across runs. Hidden states are cached as bfloat16 tensors; SAE encoding reads from the dense residual cache on demand, so top-20 selection, KMeans, and permutation null are rerunnable without re-invoking either DLM. The permutation null uses 1000 shuffles per cell with a fixed permutation seed offset, so silhouette p-values are themselves reproducible.

E Single-feature vs. Raw-probe AUC

We compare the cross-validated AUC of (i) the single most-fail-enriched SAE feature, (ii) top-N SAE features (logistic regression with per-fold selection), and (iii) a raw-probe baseline (PCA-64 + LR on the same hidden state) across all 12 cells (Figure 9). Raw probing wins in 10 of 12 cells; the two SAE wins (Qwen / MBPP, Dream / ARC) are cells where the raw probe is itself near chance. Top-20 SAE features recover roughly 85–95% of the raw-probe AUC, suggesting sparse atoms carry most but not all of the signal in the dense residual. This complements the negative steering finding (§4): even at optimal feature selection, sparse linear features under-represent the information that determines correctness.

F Dense Temporal Sweep on MBPP

We re-generate MBPP on both DLMs and capture hidden states at 9 dense checkpoints in $[48, 80]$ (every 4 steps) and an additional 6 checkpoints in $[84, 124]$ (every 8 steps), then run the same TopK-SAE encoding, top-20 fail enrichment, KMeans silhouette, and permutation null pipeline as the main 7-point grid. This sweep was requested as a robustness check against sparse-sampling artefacts. Generation uses seed 0 throughout; counts are LLaDA $115/142$ and Dream $125/132$ pass/fail. Three seeds confirm stability (Appendix I). Figure 3 plots the signal-to-null gap per step for both DLMs across the $[48, 80]$ plateau region; Appendix I reports the $[84, 124]$ late-plateau extension across three seeds. Four findings emerge. **Plateau structure.** On LLaDA-MBPP the gap stays above $+0.19$ throughout $[48, 116]$ with 6–9 sub-steps of 14 reaching $p < 0.05$ per seed. Aggregating the 14 plateau sub-steps via Fisher's combined probability test gives $p = 5.99 \times 10^{-10}$ (seed 0), 9.31×10^{-8} (seed 1), and 1.22×10^{-8} (seed 2); Stouffer's Z combination is even sharper ($Z \geq 6.09$, $p < 10^{-9}$ on all seeds). Step 64 (the only $p < 0.05$ point on the sparse 7-grid) sits in the middle of this plateau rather than at a sharp peak, so the sparse-grid claim that step 64 is the unique peak should be read as "step 64 is a representative point in the plateau". **Late collapse.** The signal collapses sharply at step 124 ($\sim$3 steps before the final state), consistent with the residual stream re-organising as the generation completes block-by-block commits. **Plateau feature hand-off.** `f15601` (LLaDA) is the top-1 fail-enriched feature at most dense steps in $[48, 100]$ and is replaced by `f3892` at the late-plateau steps 108–116. Dream-MBPP also shows a persistent top-1 marker (`f6326`) over $[60, 76]$, so a similar within-plateau hand-off behaviour appears across DLMs even when the plateau itself is weaker. **Cross-DLM concentration.** Dream-MBPP shows the same temporal shape (a flat region across the middle of denoising) but at a much lower concentration: the gap never exceeds $+0.16$ and no sub-step reaches $p < 0.05$ under the same permutation null. As noted in §4, this makes LLaDA $\times$ MBPP a

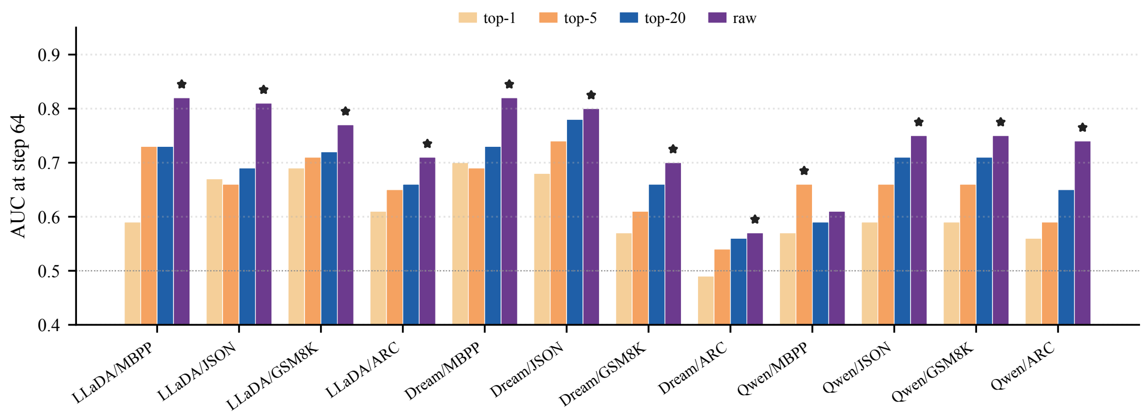

Figure 9: Per-cell 5-fold cross-validated AUC at step 64 predicting functional correctness from top-N SAE features (logistic regression with per-fold top-N selection) versus a raw-probe baseline (PCA-64 + LR on the same SAE-layer hidden state). A star marks the per-cell winner; raw probe wins in 10 of 12 cells.

model-specific point rather than a generic property of code generation, consistent with LLaDA's block-wise unmasking committing the function-signature block early and exposing fail-mode tokens at a denoising phase when later positions remain uncertain. We return to this contrast in §4.

G Cross-Layer SAE Specificity

To check that our choice of L26 is empirically motivated rather than an arbitrary single-layer restriction, we re-run the silhouette-with-permutation-null pipeline on cached LLaDA-MBPP plateau residuals at SAE layers $\{11, 16, 26, 30\}$ (the available DLM-Scope LLaDA Mask-SAE checkpoints) at steps 64 and 68. Figure 5 shows that L26 is the unique peak: step 68 at L26 reaches gap $+0.357$ with permutation $p = 0.006$, while L11, L16, and L30 give gaps below $+0.21$ and none reach $p < 0.05$. Moreover `f15601` is the top-1 fail-enriched feature only at L26 (different layers select different atoms: f13087/f541 at L11, f2741/f15654 at L16, f8020/f654 at L30), so the persistent-feature claim is itself L26-specific.

Multi-layer steering control. We further repeat the suppression protocol of §4 at the same three non-L26 layers, using each layer's top-1 fail-enriched feature ($\alpha = 5$, $t \geq 64$) on $n = 60$ samples per layer (30 cluster-1 fails, 10 cluster-0 fails, 20 pass). All three layers yield $0/60$ flips in every condition, matching the L26 within-plateau null at $0/202$. The within-plateau steering null therefore generalises across the layer span; the strongest within-plateau signal (L26) is not a controllable knob and weaker layers fare no better.

H Aggregated Significance Across Cells

The Holm-Bonferroni correction on the 40 sparse-grid tests retains 0 cells (§4), but the sparse 5-step grid hides a second aggregated-significance cell. For each of the 8 (DLM, task) cells we combine the five sparse-step p-values via Fisher's combined probability test (clamping $p_{\min} = 10^{-3}$ to avoid $\log 0$). Figure 4 shows the result: *Dream-JSON* reaches Fisher $p = 0.047$, a second cell beyond the LLaDA-MBPP dense plateau where the within-cell trajectory exceeds chance once we look at the full trajectory rather than any single step. This corroborates the cross-DLM phase-alignment claim: code/structural tasks have aggregated significance on both DLMs (LLaDA-MBPP via dense sweep, Dream-JSON via 5-step aggregation), while reasoning tasks have aggregated significance on neither.

I Seed Sensitivity

We re-run the LLaDA-MBPP dense sweep (Appendix F) with two additional generation seeds. Pass/fail counts vary by at most ± 3 samples across seeds (seed 0: $115/142$, seed 1: $112/145$, seed 2: $116/141$). The plateau structure is preserved: each seed reaches $p < 0.05$ at ≥ 7 of 9 dense sub-steps in $[48, 80]$, with `f15601` as the majority top-1 fail-enriched feature across 48–80 and `f3892` taking over at 108–116. The 3-seed gap-versus-step overlay is shown in Figure 1; per-seed numerical traces (gap, p-value, top-1 feature at every checkpoint) are released with the code.

J Intervention Effect Size

Across all n=202 samples per condition, the SAE-layer residual perturbation reaches

$\|\Delta h\|/\|h\|$=4.4% for single-feature suppression of f15601 at s64 and s16 ($\|\Delta h\|$=185, a_{target}≈9.43), 14.8% for the top-5 combined suppression at s64 ($\|\Delta h\|$=624, a_{target}=47.4), and 4.4% for the reverse intervention at s64 ($\alpha = -5$). The target feature is reliably active during every intervention window, so the within-plateau null is consistent with functional correctness being distributively encoded: perturbing a sparse direction at the SAE-layer residual does not propagate into a label flip at step 64 across 202 samples, even when that perturbation reaches ∼15% of the residual norm. Only intervention at step 16, before the plateau forms, produces any flips (3/202); the marginal effect concentrates on cluster-1 cases where f15601 fires most strongly, suggesting a weak upstream causal contribution that does not survive once the plateau features are established.

Counterfactual feature replacement. A stronger protocol simultaneously subtracts the top-5 fail-enriched directions and adds the top-5 pass-enriched directions at L26 from step 64 (replacing rather than merely ablating the fail cluster signature; top pass-enriched features at step 64 are f7961, f9732, f3011, f4420, f11861). Across the same $n = 202$ samples, this counterfactual yields $0/152$ fail→pass conversions on cluster-1 and cluster-0 cases combined, and a net-zero 1 fail→pass / 1 pass→fail on the 50 pass controls. Pairing the fail-out and pass-in directions therefore does not unlock the controllability that plain suppression failed to demonstrate.

Protocol sweep. Concerned that the negative result might reflect one too-narrow protocol, we additionally sweep 17 single-layer linear configurations at L26 with $n = 15$ cluster-1 fails and $n = 5$ pass controls each: magnitude $\alpha \in \{1, 5, 10, 20, 50\}$ on f15601 at $s64$ (perturbation norm $\|\Delta h\|$ scales from 69 to 3152); start step $\in \{0, 4, 32, 48\}$ at $\alpha = 5$; reverse $\alpha \in \{-5, -10, -20\}$ amplifying f15601; multi-feature suppression of the top-5 and top-10 fail features; and counterfactual replacement at $K \in \{1, 5\}$ including a strong $\alpha = 20$ variant. Every one of the 17 protocols yields $0/15$ fail→pass on cluster-1, for an aggregate $0/255$ across the entire single-layer linear sweep on *sparse* SAE directions.

Raw-residual directional steering. The 5–15% of correctness information not captured by the top-20 sparse subspace (Figure 9) raises a natural question: would steering along a *raw* residual direction, rather than a sparse SAE feature, succeed where sparse steering fails? We test three candidate raw directions at $L26$ step 64: (i) the pass-minus-fail mean-difference vector ($\bar{h}_{\text{pass}} - \bar{h}_{\text{fail}}$, unit-normalised), (ii) the top principal component of the residual covariance oriented toward pass, and (iii) a Fisher LDA direction with regularised within-class covariance. Across 6 configurations (mean-diff at $\alpha \in \{5, 20, 50\}$, PCA-1 at $\alpha = 5$, Fisher at $\alpha = 5$, and mean-diff at $\alpha = 5$ from step 16), every protocol yields $0/15$ fail→pass on cluster-1 fails, for an aggregate $0/90$ on top of the SAE sweep. The largest perturbation ($\alpha = 50$, raw mean-diff) drives $\|\Delta h\|$ to 71,168 (∼17× the residual norm), well beyond the magnitudes used in successful AR steering studies, and still produces zero flips. The combined linear-steering ledger at L26 thus stands at $0/345$ fail→pass over 23 protocols. This rules out the hypothesis that the sparse-subspace restriction is the culprit (raw directions occupy the full residual and still fail), and tightens the remaining explanation toward DLM-specific structural causes such as block-commit timing or non-linear computational paths. Gradient-based, multi-layer joint, and non-linear interventions remain open.

K Post-hoc Selection Sensitivity

We reanalyse cached LLaDA-MBPP plateau residuals (seed 0) at the strongest plateau steps and report three robustness checks. **(a) Top-N.** Figure 10 plots the signal-to-null gap as N varies over $\{10, 20, 30, 50\}$. Step 68 remains $p < 0.05$ at every N, while step 64 degrades for $N \geq 30$, marking step 68 as the more robust plateau point. **(b) Clustering method.** At N=20, KMeans, Agglomerative (Ward), and HDBSCAN ($\min\_\text{size}$=5) all recover K=2 with near-identical silhouette at both steps (0.609 / 0.719 for step 64 / 68; HDBSCAN drops 11 samples as noise). **(c) Labels-hidden.** Clustering the full 257-sample fail+pass set at K=2 with the same top-20 features (selected via labels, cluster unsupervised) gives ARI ≈ -0.007 at both steps ($p > 0.85$), confirming the silhouette signal measures within-fail sub-structure rather than fail/pass separability. **(d) Held-out feature selection.** To address selection-bias concerns even after the permutation null already re-selects features inside each shuffle, we partition the 142 fail samples into 5 stratified folds, select the top-20 fail-enriched features on the held-in 4 folds, and

feature	dominant context (F/P split)	top quotation
`f15601`	complex-math docstring (25/5)	*"nth octagonal number...O(n)=(n^2+3n)//2"*
`f3892`	`:param`/`:return` annotation (15/15)	"`:param tuple_list:` *input list of tuples*"
`f5561`	arithmetic variable assignment (22/8)	*"max_product = min_product"*
`f11265`	broken-syntax / incomplete code (17/13)	*"dp[n == 0] = [1 \n dp = [0]"*
`f2144`	docstring opening newline (14/16)	*"def fn(x):\n """\n"*

Table 6: Activation-maximising contexts for the top fail-enriched features on LLaDA-8B / MBPP at step 64; fail/pass counts are among each feature's top-30 examples. `f15601` fires on complex-math docstring contexts (task-difficulty correlate), `f11265` fires on broken-syntax contexts (closer to a true failure detector), and `f2144` fires on a structural position (non-discriminative).

compute the KMeans silhouette on the held-out fold (mean across folds). Across the nine dense plateau steps in $[48, 80]$, the held-out silhouette is 0.63–0.73 with held-out null mean 0.37–0.42 (gap $+0.25$ to $+0.34$); under a 500-shuffle permutation null using the same held-out pipeline, *all nine steps reach* $p < 0.05$ (best $p < 10^{-3}$ at steps 56/68/72), which is stronger than the $7/9$ achieved by in-sample selection. The plateau is therefore not an artefact of label-driven feature selection.

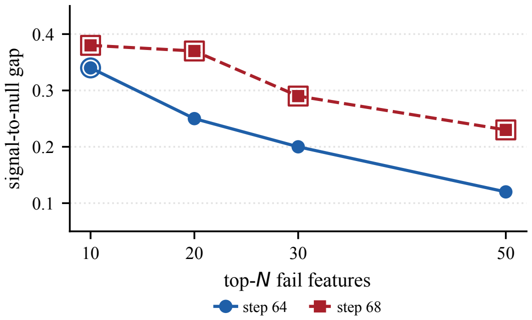

Figure 10: Signal-to-null gap as the top-N fail-feature cutoff is varied at LLaDA-MBPP plateau steps 64 and 68. Hollow rings mark $p < 0.05$. Step 68 is robust across N; step 64 only reaches significance at N=10.

L Activation-Maximising Examples

To turn the statistically discovered top features into interpretable failure markers, we re-run LLaDA-8B at step 64 on all 257 MBPP samples, encode every generation position through the L26 Mask-SAE, and record the top-30 *(sample, position)* entries per target feature. Table 6 summarises the dominant context type for each feature, the fail-rate in its top-30 examples, and a representative quotation.

These contexts sharpen the reading in §4. `f15601`'s top activations are dense mathematical specifications (Eulerian / octagonal / hexagonal / Babylonian formulae) where the model is more likely to fail, so ablating this task-difficulty signal does not flip correctness because the perturbation is upstream of the failure mechanism, consistent with the within-plateau steering null (§4). By contrast `f11265` fires on contexts that already contain visible syntactic defects, closer to a genuine failure detector than the mid-plateau atoms, while `f2144` fires on a purely structural position and is non-discriminative.

M Functional Correctness Definitions

Every task uses a deterministic, machine-checkable rubric; samples with no extractable candidate are labelled fail. **MBPP:** extract the first `def`-block or fenced ```` ```python ```` block, prepend `test_imports`, run all assertions with a 10-second timeout; syntax errors, timeouts, and assertion failures are fails. **JSON schema:** output must parse as valid JSON and match the reference structure (top-level type, required keys, scalar-leaf types), ignoring key order and extra keys. **GSM8K:** compare only the final numeric answer; extract `####\s*([-\d.]+)` and compare to gold after stripping commas and trailing zeros. **ARC-Challenge:** extract `####\s*([A-D])`; fallback to the last standalone A–D letter.

N Cross-Task Transport Replication

To test whether the cross-generator transport result generalises beyond MBPP, we repeat the protocol of §5 on JSON-schema generation. We generate Qwen-2.5-7B-Instruct solutions for all 272 JSON-schema prompts under the same chat template and functional rubric (Appendix M), and ask how well Dream-Base's per-layer hidden states predict *Qwen-Instruct*'s pass/fail versus *Dream-Instruct*'s own pass/fail (5-fold logistic CV, C=0.01, last-prompt-token state). Qwen-Instruct passes $162/272$ (59.6%); per-sample pass/fail agreement with Dream-Instruct is 0.783. The transport is if anything stronger than on MBPP: at every layer above L16 the AUC for predicting Qwen-Instruct's labels *exceeds* the AUC for Dream's own labels (best Dream layer L25: 0.744 on Dream vs 0.771 on Qwen, $\Delta = +0.027$; at L26 the gap is $+0.051$). A probe built on Dream's representation therefore predicts a non-generating model's JSON correctness at least as well as Dream's own, replicating the MBPP finding that the layer encodes shared problem difficulty rather than generation-specific correctness.

We also run the protocol on GSM8K, a reasoning task where difficulty co-evolves with the answer (Qwen-Instruct passes $1200/1319 = 91\%$, Dream-Instruct $811/1319 = 61.5\%$, agreement 0.663). The transport is again positive at every layer, and the gap is larger: the best Dream layer (L20) predicts Dream's own correctness at AUC 0.651 but Qwen-Instruct's at 0.719 ($\Delta = +0.068$). That Dream features predict a stronger non-generating model's failures *better* than Dream's own is the difficulty reading in its sharpest form: Qwen-Instruct fails GSM8K almost only on genuinely hard problems, so a prompt-difficulty probe tracks its pass/fail more cleanly than Dream's noisier self-correctness. Because Qwen-Instruct's GSM8K failures are only 9% of items, this AUC is computed on an imbalanced label set and should be read alongside the more balanced MBPP and JSON cells.

To triangulate the shared difficulty ordering with a third, architecturally unrelated generator, we repeat the test with Llama-3.1-8B-Instruct. Dream's features predict Llama-Instruct's pass/fail almost as well as Dream's own: on MBPP, AUC 0.762 for Llama vs 0.771 for Dream at the best layer ($\Delta = -0.009$, Llama pass $176/257$, agreement 0.673); on JSON, 0.726 vs 0.744 ($\Delta = -0.018$, Llama pass $128/272$, agreement 0.761). A third generator from a different model family therefore transports with < 0.02 AUC loss, so the L26 signal indexes generator-agnostic problem difficulty rather than a property of the Dream–Qwen pair.

Finally, we test a held-out code dataset never used elsewhere in the pipeline, HumanEval (164 problems). Dream-Base features (a forward-pass difficulty estimate, no DLM generation) predict the pass/fail of two independent AR generators on the same problems: Qwen-Instruct at AUC 0.66 and Llama-Instruct at 0.69 (mid-layer), with the two generators agreeing 0.689. The AUC is more modest than on MBPP/JSON, as expected for a dataset Dream never trained on, but the difficulty transport, a DLM representation predicting independent generators' correctness, holds on a new code task.

O Probe-Class and Regularization Robustness

Because the difficulty reading rests on probe behaviour, we check that it is not an artifact of the probe family or its regularization (Dream, MBPP, per-step best layer). Replacing the linear logistic probe with an RBF-kernel SVM changes peak AUC by at most ± 0.01 at every checkpoint (step 16: 0.802 vs 0.804; step 32: 0.830 vs 0.824; step 64: 0.837 vs 0.827; step 127: 0.876 vs 0.878), so a non-linear probe extracts no additional separability and the signal is essentially linearly accessible. Nested cross-validation, which selects layer and regularization on an inner fold and evaluates on a held-out outer fold, lowers AUC by only 0.021 on average (0.038 worst-case), confirming the per-layer numbers are not the product of regularization tuning. Per-fold AUC standard deviation at step 64 is 0.03.

P Difficulty Stratification

As a second disentanglement instrument complementing cross-generator transport (§5), we stratify by a per-prompt difficulty proxy. A probe trained to predict prompt length alone reaches AUC 0.78 at step 64, confirming that an explicit difficulty surrogate is itself linearly decodable from the residual. Re-fitting the correctness probe within prompt-length-matched strata (n=196) lowers AUC by 0.026–0.036 across steps (step 0: $0.757 \to 0.721$; step 16: $0.816 \to 0.789$; step 64: $0.835 \to 0.809$; step 127: $0.877 \to 0.849$). Length-matching therefore removes a measurable but minor part of the signal: prompt length is one crude difficulty axis, whereas the cross-generator transport (which retains 94% of AUC) captures shared difficulty more completely. The two instruments agree in direction: a substantial fraction of what the correctness probe reads is a prompt property, not a generation-specific one.

Q Alternative-SAE Robustness

To check that the plateau signal is not an artifact of one SAE dictionary, we re-encode the LLaDA-MBPP step-64 residuals at L26 through three DLM-Scope dictionaries of increasing density (trainers 0/2/4, $k = 50/160/520$) and recompute the top-20 fail-enrichment, KMeans silhouette, and permutation null. The within-fail cluster structure is significant for the two higher-capacity dictionaries (gap $+0.294$, $p = 0.038$ at $k = 160$; gap $+0.313$, $p = 0.022$ at $k = 520$) and weaker only at the very sparse $k = 50$ (gap $+0.147$, $p = 0.134$), so the plateau is not specific to the $k = 160$ SAE used in the main text. The dominant fail feature `f15601` is top-1 at $k = 50$ and $k = 160$; at $k = 520$ the denser dictionary splits the same difficulty direction across atoms (top-1 becomes `f11355`), so the *phenomenon* is dictionary-robust while the exact atom identity is sparsity-dependent, as expected for SAEs.

R Denoising-Progress Alignment

The cross-model step-axis comparison (Figure 2) pairs LLaDA's block-wise schedule with Dream's global schedule at matched checkpoint steps. Although the unmasking *order* differs, the unmasked *fraction* is nearly identical at every checkpoint, so the comparison is aligned in denoising progress rather than only loosely:

step	0	1	4	16	32	64	127
LLaDA (block)	0%	0.8%	3.1%	12.5%	25%	50%	99%
Dream (global)	0.8%	1.6%	3.9%	13.3%	26%	51%	100%

The fractions are deterministic from each schedule (LLaDA commits 32-token blocks over 16 steps each; Dream unmasks on a global linear timestep), and differ by $\leq 1.6\%$ at every checkpoint. At the plateau step 64 both models have committed $\sim 50\%$ of the generation region, so the mid-denoising phase compared across models corresponds to matched denoising progress.

S Gradient Attribution of the Diagnostic Features

The steering null shows the diagnostic features cannot be *driven* to change outcomes; gradient attribution asks the complementary question without intervention: do they lie on a salient path from the L26 residual to the output? At the step-64 partially-denoised state we cut the computation graph at L26 (insert the residual as a leaf), backpropagate the model's self-confidence (sum of $\log p$ of its argmax over masked generation positions), and score each SAE feature by $\left(\nabla_h \cdot W_{\text{dec}}[:, i]\right) z_i$, the gradient-times-activation in feature space, averaged over 20 fail samples. Of the 491 features active across these samples, the diagnostic features rank low: `f15601` ranks 220th (attribution 0.13% of the maximum), `f3892` 55th, and `f11265` 62nd. The top-attribution features (`f5978`, `f3689`, `f12026`, ...) are disjoint from the fail-enriched set. Independently of intervention, then, the difficulty features are not on a salient gradient path to the output, corroborating the diagnostic-not-causal reading.

T Pooling Granularity

Region-mean pooling over the four generation regions could in principle manufacture the plateau cluster structure by averaging. Re-running the L26 step-64 diagnose on each region separately, `f15601` remains the most fail-enriched feature in the docstring-bearing first region (enrichment 0.39), consistent with its difficulty reading, and regions 1–3 show positive cluster gaps ($+0.15$ to $+0.19$). No single region reaches permutation significance on its own (p between 0.13 and 0.81); the region-mean aggregates the consistent per-region enrichment into the significant $+0.29$ ($p = 0.038$). Pooling therefore aggregates a distributed weak signal rather than fabricating one from noise, though the per-step significance does rely on this aggregation, which we note as a sensitivity of the pooling choice.